STUDYMATE • MILESTONE SUMMARY

Backend Engineering & Retrieval Pipeline

Core Architecture, Processing Workflows & RAG Integration

MILESTONE OVERVIEW

This milestone focused on building and integrating the essential backend components that power the system's main operations. It included developing core features such as PDF text extraction, chunking, semantic embedding generation, and similarity-based retrieval using FAISS. These functionalities formed the backbone of StudyMate, enabling accurate question-answering and efficient information retrieval from academic documents.

CORE BACKEND DELIVERABLES

COMPONENT	TECHNICAL FUNCTIONALITY & IMPACT
PDF Text Extraction	Parses complex academic document formats to accurately isolate structured textual data.
Document Chunking	Splits raw text into optimized contextual windows to prepare data for embedding models.
Semantic Embeddings	Generates high-dimensional vector representations to capture deep semantic context.
FAISS Vector Search	Executes similarity-based retrieval to deliver rapid, highly accurate document context for Q&A.